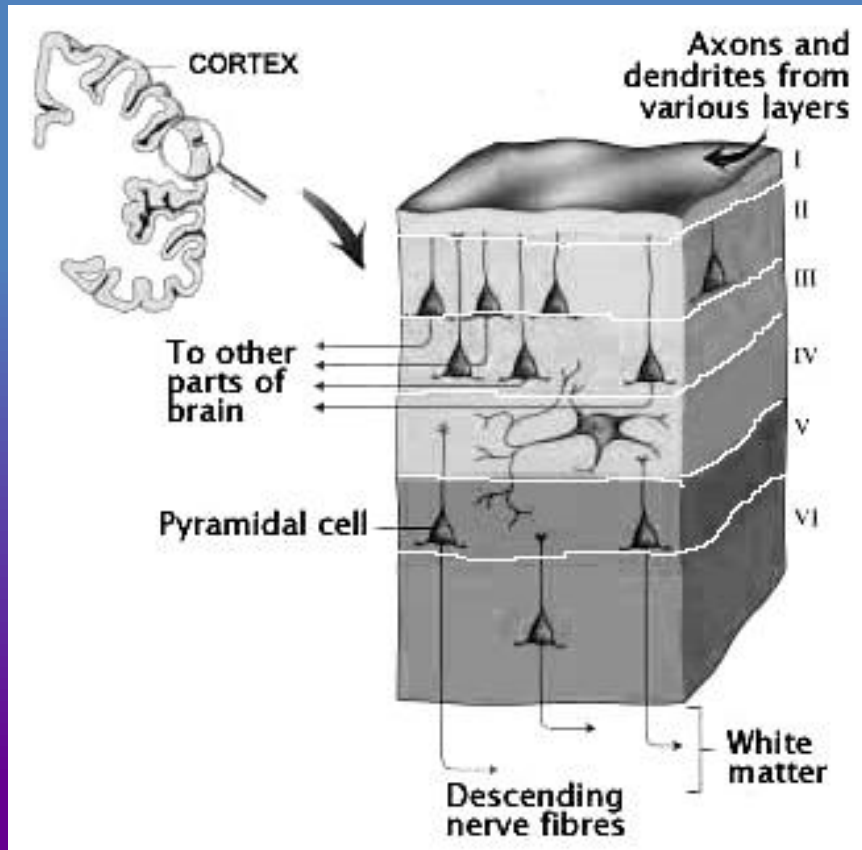
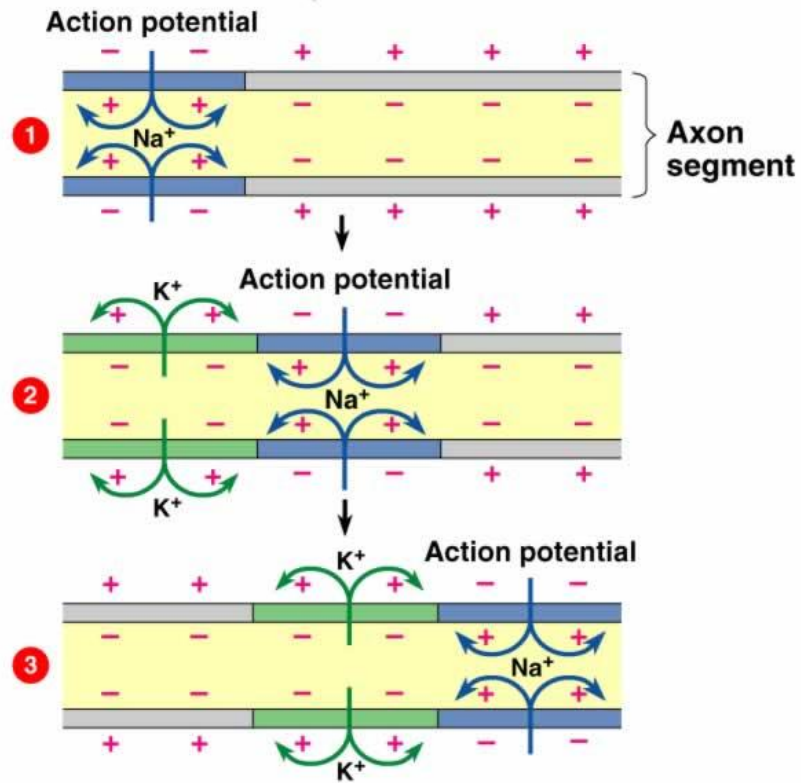
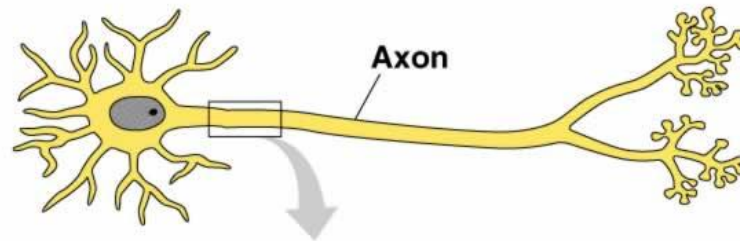


Neocortical Layers



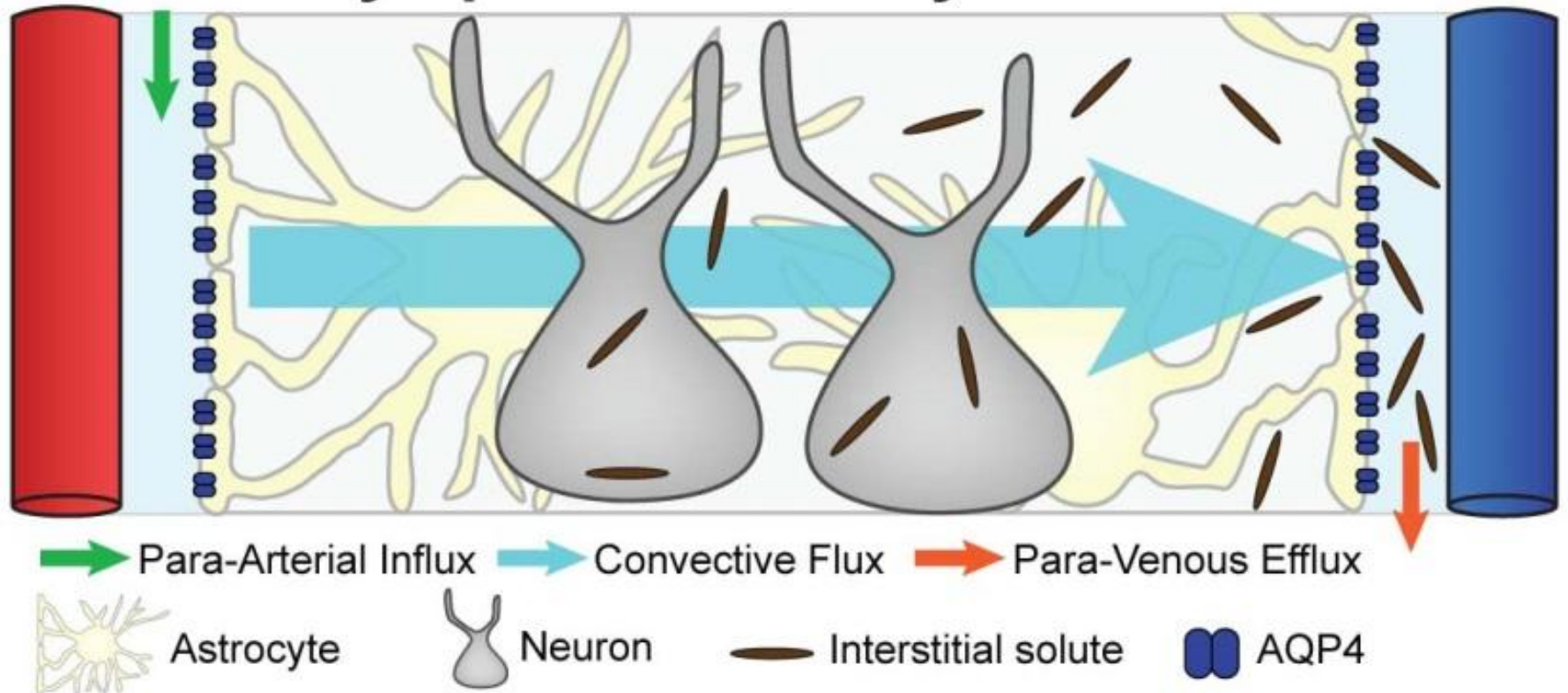


Glymphatics

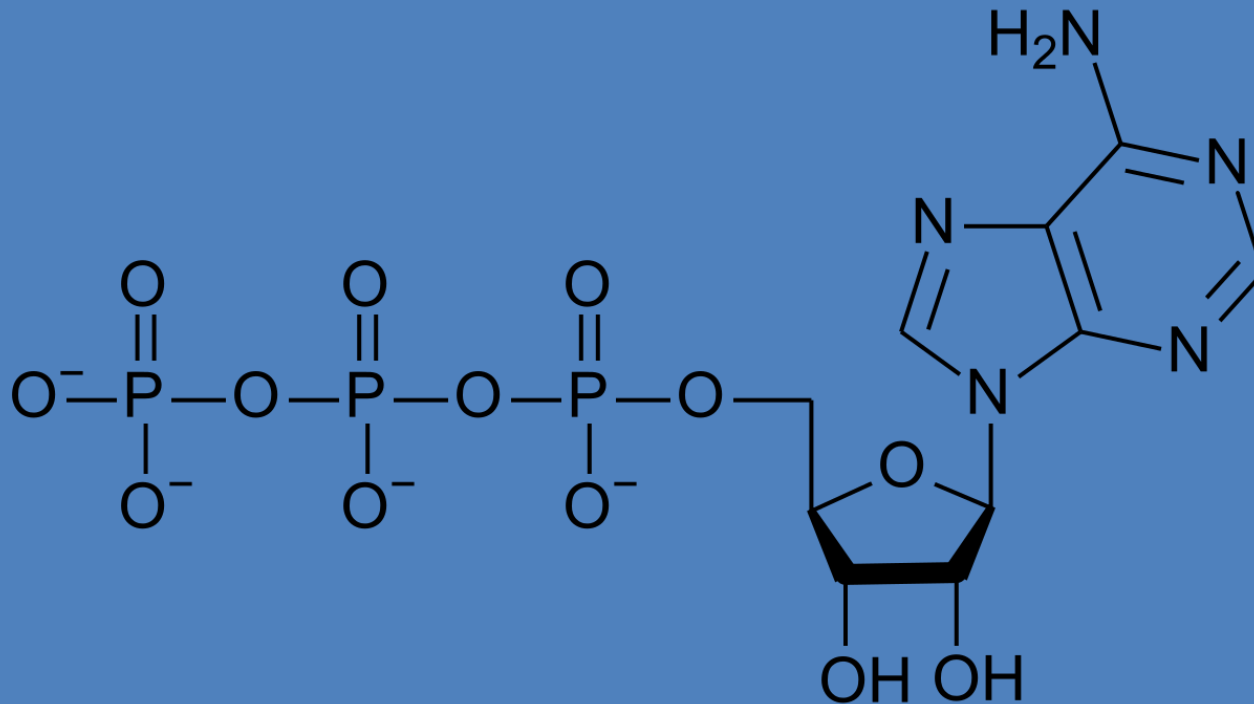


Does sleeping wash our brains?

Glymphatic Pathway Function



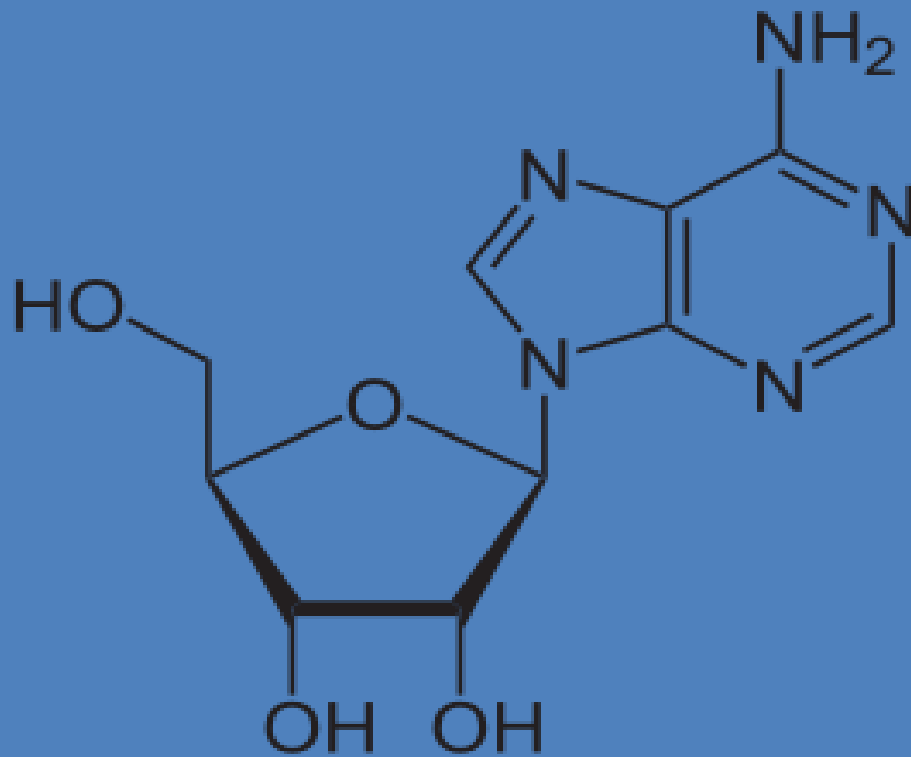
Adenosine Triphosphate ATP



Synaptic Homeostasis

- “sleep is for an energy surge”, a surge that permits energy-consuming anabolic processes, such as protein and fatty acid synthesis, to occur.
- Higher energy utilization during wake and its reversal during sleep, since increased synaptic potentiation during wake would contribute to high energy consumption and the surge in ATP during sleep would reflect the energy conserving processes such as synaptic downscaling during sleep.
- The availability of ATP in sleep could promote the changes related to synaptic plasticity and memory consolidation and remodeling during sleep

Adenosine



A Teenager Died After Drinking Too Much Caffeine

Drank a diet Mountain Dew, a cafe latte from McDonald's, and an energy drink within two hours, a coroner ruled
BuzzFeed May 16, 2017.



54 mg in can



77 mg per
shot



160 mg per 16 oz can

USA Today

Family blames popular energy drink for teen's death, lawsuit says

A family is suing a distributor of Alani Nu energy drinks. The lawsuit blames energy drinks for her death and says they need better...

5 days ago

People.com

Texas Teen Died Due to 'Enlarged Heart' from Caffeine from Energy Drink, Attorney Claims: Reports

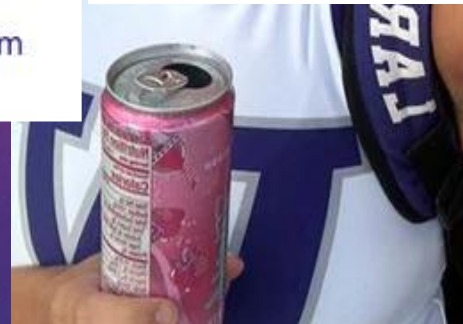
The family of Texas teenager Larissa Rodriguez, who died in October 2025, filed a wrongful death lawsuit against the distributors of Alani...

6 days ago

WOAI

Teen dies after drinking 'a large amount of caffeine' from energy drinks

Cardiomyopathy attributed to stress and excessive caffeine



At least 1 can daily for months

10/20/25

MARKETING OF 300MG CAFFEINE TO CHILDREN \$2 a can



Can sleepiness be measured?



Epworth Sleepiness Scale

Use the following scale to choose the **most appropriate number** for each situation:

- 0 = would **never** doze
- 1 = **slight chance** of dozing
- 2 = **moderate chance** of dozing
- 3 = **high chance** of dozing

It is important that you answer each question as best you can.

Situation	Chance of Dozing (0-3)
Sitting and reading _____	_____
Watching TV _____	_____
Sitting, inactive in a public place (e.g. a theatre or a meeting) _____	_____
As a passenger in a car for an hour without a break _____	_____
Lying down to rest in the afternoon when circumstances permit _____	_____
Sitting and talking to someone _____	_____
Sitting quietly after a lunch without alcohol _____	_____
In a car, while stopped for a few minutes in the traffic _____	_____
TOTAL	_____

Multiple Sleep Latency Test

- MSLT invented at Stanford and now used worldwide!
- Standardized test to measure sleepiness
- Consists of five 20- nap opportunities during the time expected to be awake
- Person is in bed and asked to go to sleep
- If they fall asleep within 20 minutes the nap is extended for 15 minutes to see if REM sleep occurs (SOREM)
- Average time to fall asleep is reported as the sleep latency, max is 20 minutes and minimum is zero.
- Scores of 8 minutes or less consistent with EDS





Obstructive Sleep Apnea

Rafael Pelayo, MD



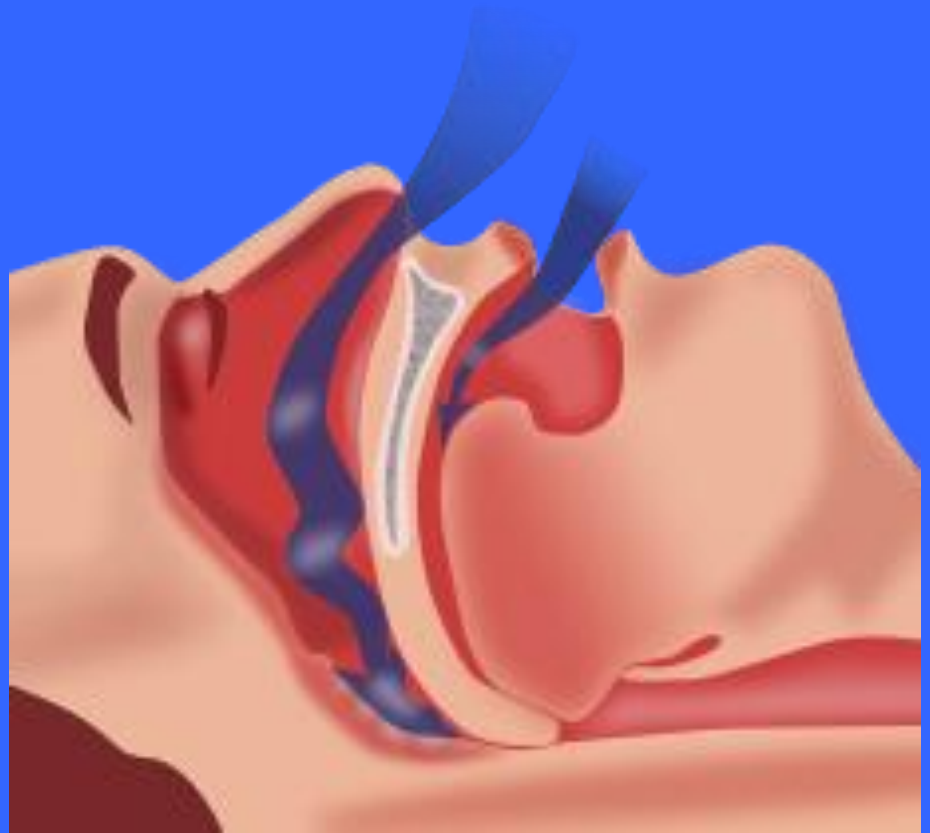
STANFORD
SCHOOL OF MEDICINE

Sleeping should be silent



Obstructive Sleep Apnea

Intermittent pauses or difficulty breathing while sleeping due to a blockage in the throat

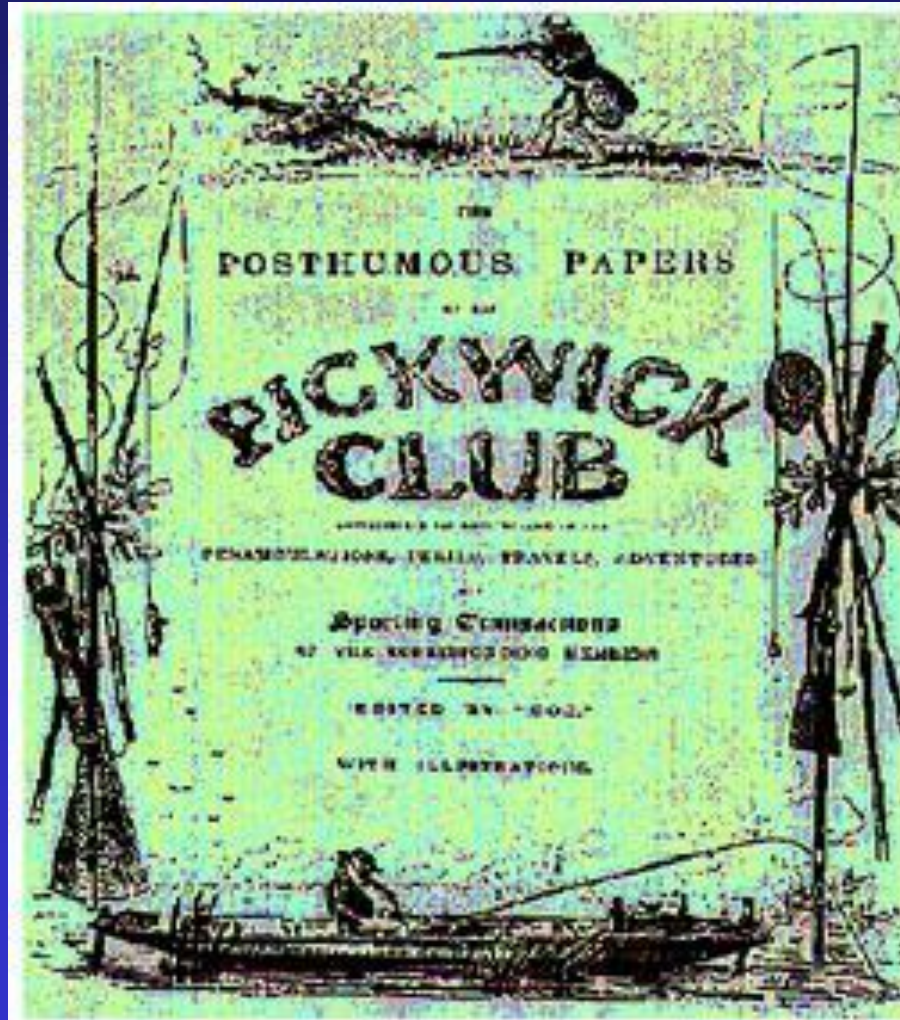


The brain prioritizes sleeping over breathing!

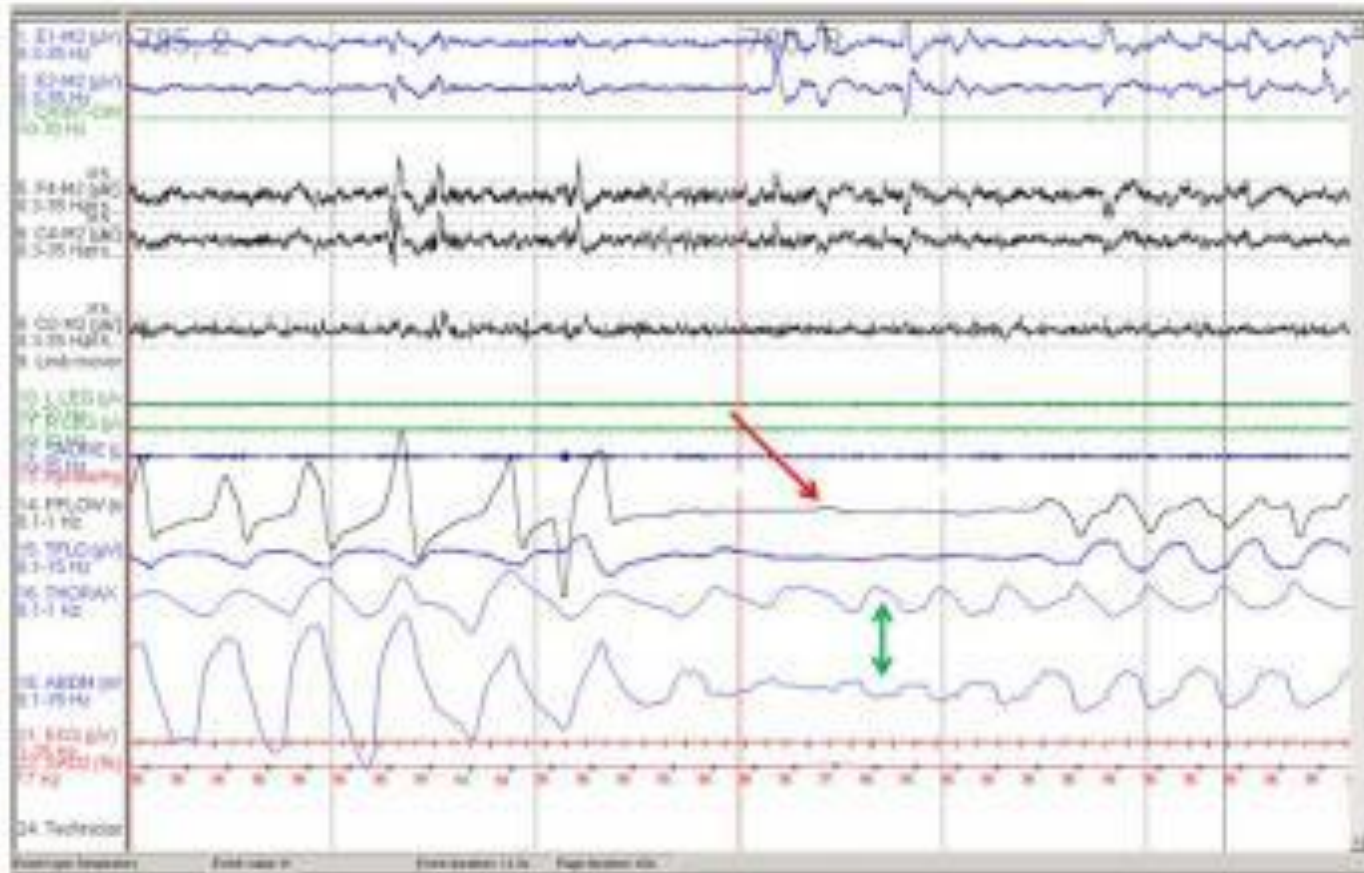


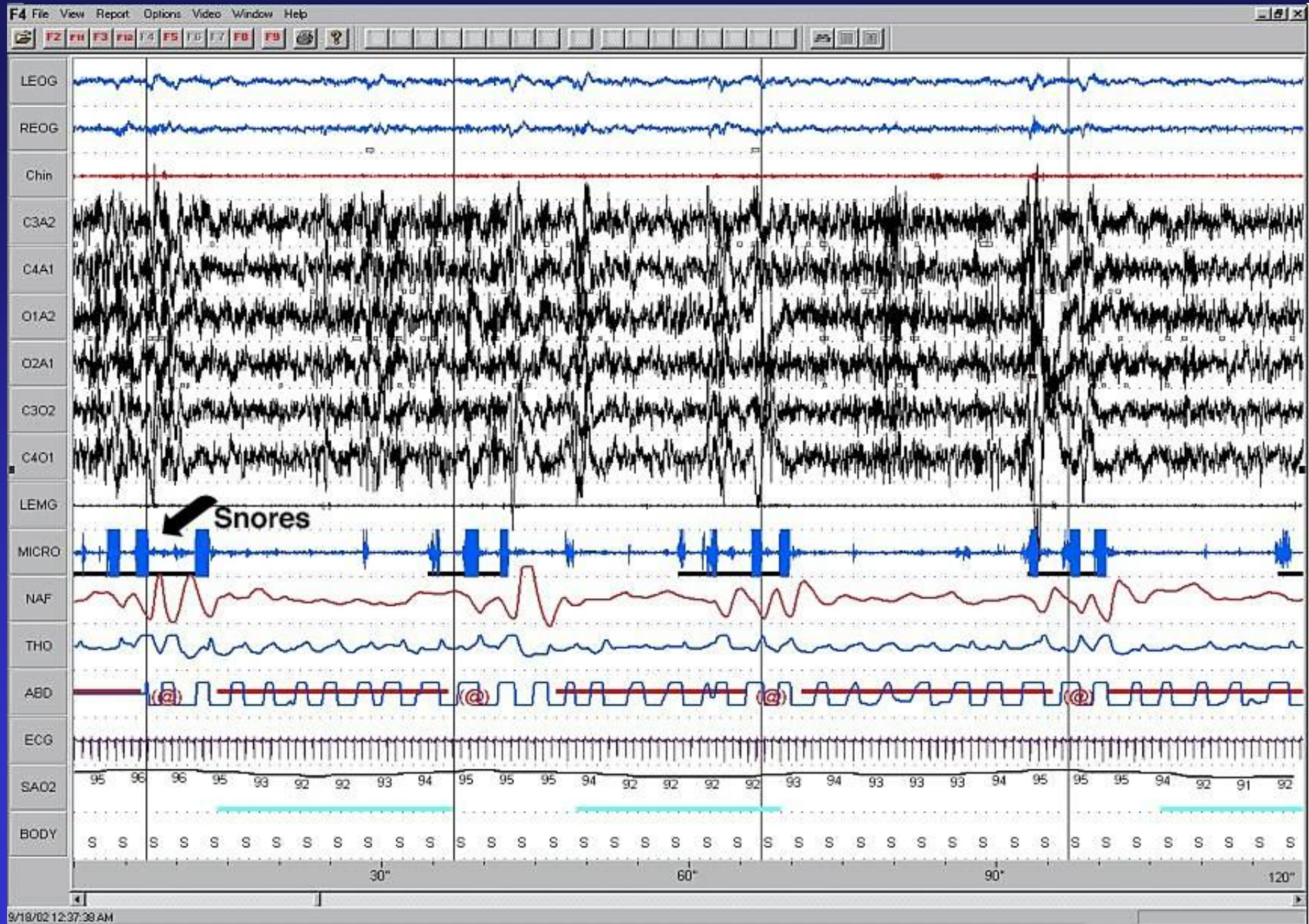
“...and on the box sat a fat and red-faced boy, in the state of somnolency.” C. Dickens

Pickwickian Syndrome



Obstructive Sleep Apnea





Obstructive Sleep Apnea

- Apnea is a pause in breathing, usually 10 seconds or longer
- Hypopnea is a decreased airflow or series of partial breaths that disrupt sleep or cause a decrease in oxygenation
- Sleeping is repetitively for brief time periods more important than breathing!

OSA Features

- Hypertension
- GI Reflux
- Morning headaches
- Unrefreshing sleep
- Behavioral awake problems
- Obesity (but thin people can have OSA too!)
- Excessive sleepiness
- Insomnia
- Bed wetting/ secondary nocturnal enuresis
- Night sweats
- Snoring/ Mouth breathing

OSA MORBIDITY

- OSA is a risk factor for cardiovascular disease and mortality. linked to hypertension, stroke, ischemic heart disease, cardiac arrhythmia, and pulmonary hypertension. OSA can cause endothelial dysfunction and vasculature remodeling, which can lead to atherogenesis. OSA can increase the risk of sudden death.
- OSA is associated with an increased risk of cancer. OSA is associated with metabolic, cardiovascular, and cerebrovascular diseases.
- OSA can cause excessive daytime sleepiness (EDS). OSA can cause neurocognitive deficits, such as impairments in concentration, memory, vigilance, and executive function. OSA can cause psychological problems, such as depression or personality changes. OSA can increase the chance of accidents.
- Treatments for OSA can improve quality of life and decrease blood pressure. Early diagnosis and treatment of OSA can help prevent cardiovascular complications

Mortality

- People with moderate-to-severe OSA have a 33% mortality rate over 14 years
- People with severe OSA have a 1.9 times higher risk of all-cause mortality
- People with severe OSA have a 2.65 times higher risk of cardiovascular mortality

STATE OF CALIFORNIA
CERTIFICATION OF VITAL RECORD

COUNTY OF SANTA CRUZ
SANTA CRUZ, CALIFORNIA

INFORMATIONAL,
"NOT A VALID DOCUMENT TO ESTABLISH IDENTITY."

AMENDED

2 OF 2

2 AMENDMENT OF MEDICAL AND HEALTH SECTION DATA-DEATH

4400 - 1251

IDENTIFICATION OF THE RECORD	1. FIRST NAME Santa Cruz	2. MIDDLE NAME M	3. LAST NAME [REDACTED]
	4. PLACE OF OCCURRENCE Santa Cruz	5. DATE OF EVENT Sept., 23, 1977	6. DATE ORIGINAL FILED 10-3-77
SPATIALLY REPORTED INFORMATION	7. INFORMATION AS REPORTED ON THE UNOFFICIALLY REGISTERED CERTIFICATE Pending		
	8. FIRST & LAST NAME OF REPORTER [REDACTED]		
	9. SPECIFY A SOURCE OF INFORMATION Yes ☒ No ☐		
	10. PLACE OF BIRTH [REDACTED]		
INFORMATION AS IT SHOULD BE ENTERED ON THE OFFICIALLY REGISTERED CERTIFICATE	11. FIRST & LAST NAME OF DECEASED Undetermined, Possibly Acute Right Ventricular Failure,		
	12. SECONDARY CAUSE OF DEATH Secondary to Pickwickian Syndrome		
	13. FIRST & LAST NAME OF PHYSICIAN OR CORONER Alfred F. Norenby		
	14. SIGNATURE OF PHYSICIAN OR CORONER Alfred F. Norenby		
DECLARATION OF CERTIFYING PHYSICIAN OR CORONER	15. I, THE CERTIFYING PHYSICIAN OR CORONER, HAVE PERSONALLY EXAMINED THE DECEASED AND AM SURE THAT THE INFORMATION REPORTED HEREON IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE.		
	16. SIGNATURE OF PHYSICIAN OR CORONER Alfred F. Norenby		
	17. TITLE OF PHYSICIAN OR CORONER Coroner		
	18. ADDRESS - OFFICE AND HOME P.O. Box 623, Santa Cruz, Ca.		
REGISTRAR'S OFFICE	19. OFFICE OF STATE OR LOCAL REGISTRAR BY: WELLS S. SHIELDS, CHIEF		20. DATE ACCEPTED NOV 29 1977
	21. STATE OF CALIFORNIA DEPARTMENT OF HEALTH - OFFICE OF THE STATE REGISTRAR OF VITAL STATISTICS FOR 10-10-77, REV. 4-7-71		

Certification of death in 1977

INFORMATIONAL,
"NOT A VALID DOCUMENT TO ESTABLISH IDENTITY"



CERTIFIED COPY OF VITAL RECORDS
STATE OF CALIFORNIA, COUNTY OF SANTA CRUZ

000302933

This is a true and exact reproduction of the document officially registered and placed on file in the OFFICE OF THE SANTA CRUZ COUNTY RECORDER.

DATE ISSUED **NOV 19 2002** BY **[Signature]**

This copy is not valid unless printed on an engraved border, bearing the seal and signature of the County Recorder.

Sean Saldivia
SEAN SALDIVIA
COUNTY RECORDER



STATE OF CALIFORNIA
CERTIFICATION OF VITAL RECORD

COUNTY OF SANTA CRUZ
SANTA CRUZ, CALIFORNIA

INFORMATIONAL,
"NOT A VALID DOCUMENT TO ESTABLISH IDENTITY."

AMENDED
2 OF 2

✓ AMENDMENT OF MEDICAL AND HEALTH SECTION DATA-DEATH 4400 - 1251

INFORMATION AS IT SHOULD BE STATED ON THE ORIGINALLY REGISTERED CERTIFICATE

23. PART 1. STATE THE CAUSE OF DEATH (A) **Undetermined, Possibly Acute Right Ventricular Failure, Secondary to Pickwickian Syndrome**

24. PART 2. UNDER STANDING CONDITIONS (B) **Extreme Obesity with Associated Cardiomegaly**

25. SPEECH ACTUAL, MANNER OF DEATH (C) **1850**

26. PLACE OF DEATH (D) **1850**

27. PLACE OF BIRTH (E) **1850**

28. SEX (F) **1850**

29. AGE (G) **1850**

30. DATE OF DEATH (H) **1850**

31. DATE OF BIRTH (I) **1850**

32. DATE OF DEATH (J) **1850**

33. DATE OF BIRTH (K) **1850**

34. DATE OF DEATH (L) **1850**

35. DATE OF BIRTH (M) **1850**

36. DATE OF DEATH (N) **1850**

37. DATE OF BIRTH (O) **1850**

38. DATE OF DEATH (P) **1850**

39. DATE OF BIRTH (Q) **1850**

40. DATE OF DEATH (R) **1850**

41. DATE OF BIRTH (S) **1850**

42. DATE OF DEATH (T) **1850**

43. DATE OF BIRTH (U) **1850**

44. DATE OF DEATH (V) **1850**

45. DATE OF BIRTH (W) **1850**

46. DATE OF DEATH (X) **1850**

47. DATE OF BIRTH (Y) **1850**

48. DATE OF DEATH (Z) **1850**

INFORMATION
AS IT SHOULD
BE STATED
ON THE
ORIGINALLY
REGISTERED
CERTIFICATE

How common is OSA?

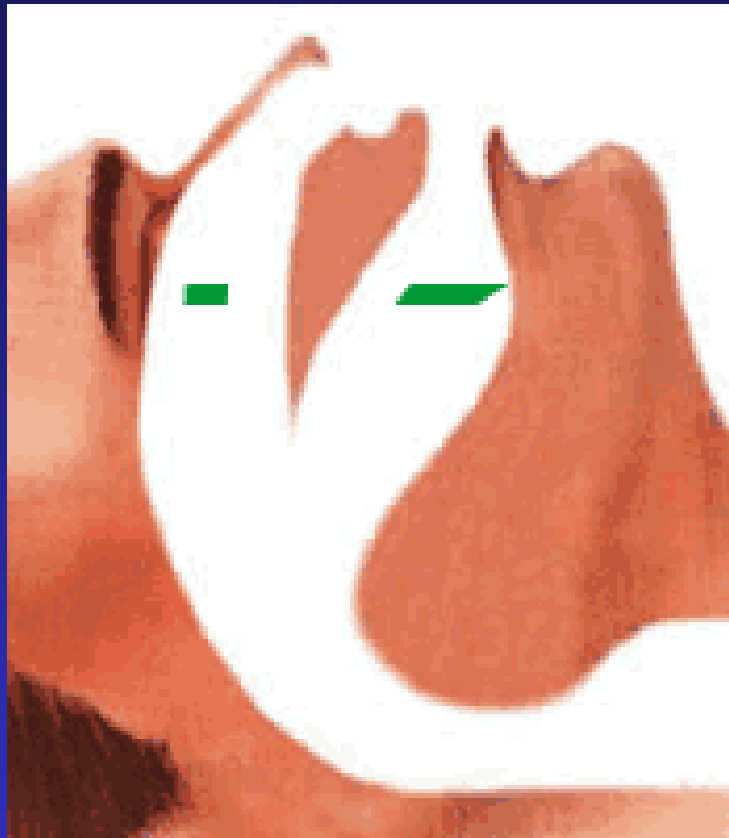
- Conservative estimated prevalence of OSA is approximately 3 to 7% for adult men and 2 to 5% for adult women in the general population, it may be as high as 26% of adults between the ages of 30 and 70 years
- Afflicts at least 25 million adults in the U.S, **1 billion world wide**
- In children estimated 1-5 %



Little coconuts do not fall far from the tree

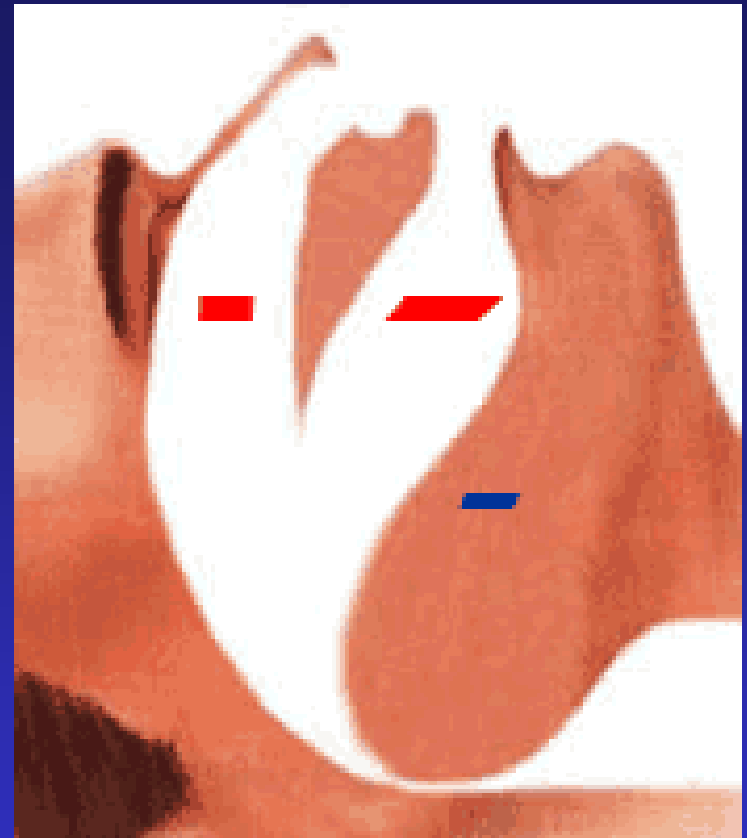
OSA complications

- Increased risk of sudden death
- Increased risk of heart attacks
- Increased risk of arrhythmias
- Increased risk of strokes
- Increased risk of car accidents
- Makes all other neurological conditions worse i.e. people with migraines get more migraines; people with seizures get more seizures



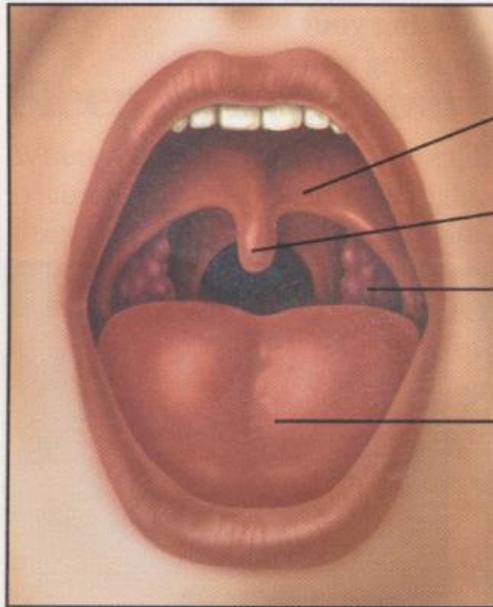
Normal Breathing

- Airway is open
- Air flows freely to lungs



Obstructive Sleep Apnea

- Airway collapses
- Blocked air flow to lungs

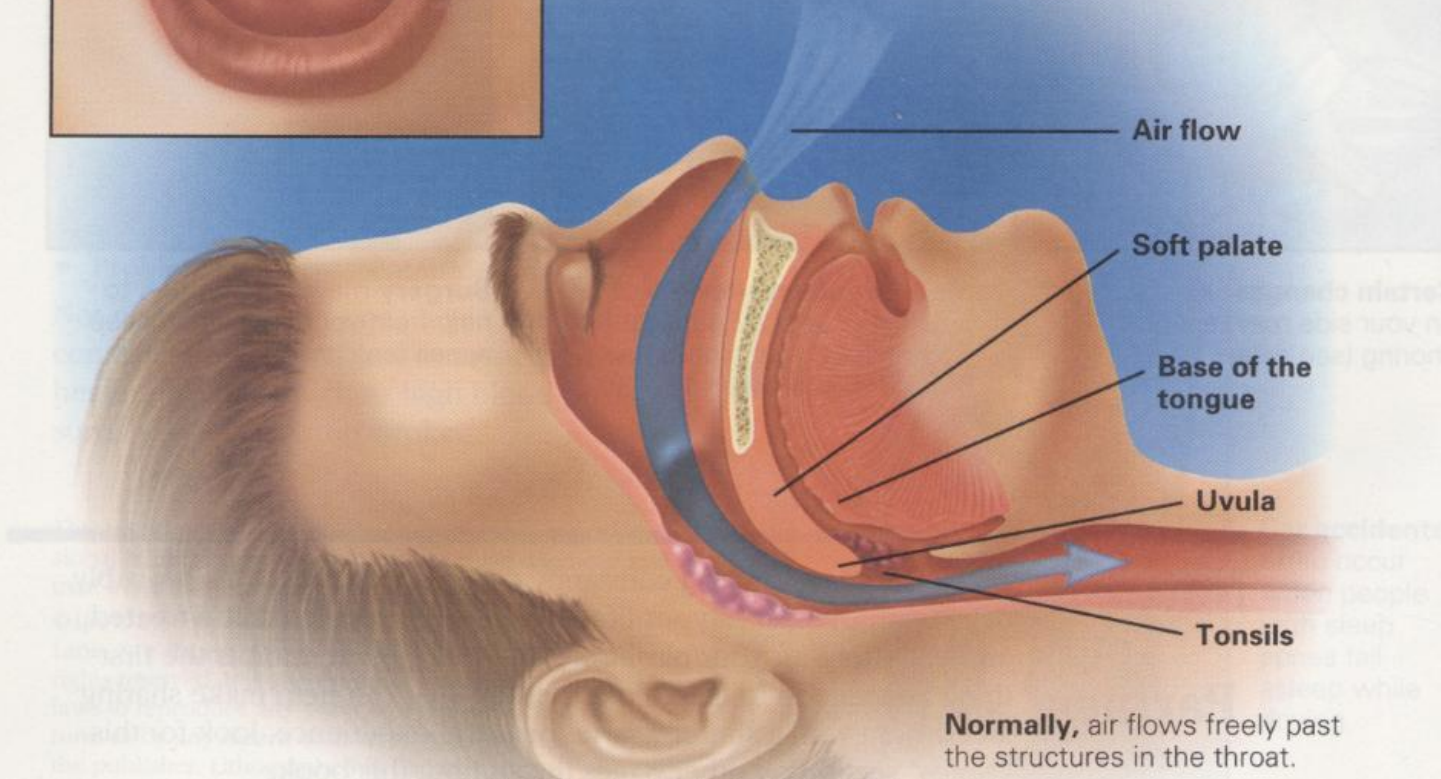


The soft palate is the tissue at the back of the roof of your mouth. It helps block off your nose when you swallow.

The uvula is a long flap of tissue that hangs from your soft palate.

Tonsils are balls of tissue in the throat. They may play a small role in helping your body defend itself against illness.

The tongue helps you talk, chew, and swallow.



Air flow

Soft palate

Base of the tongue

Uvula

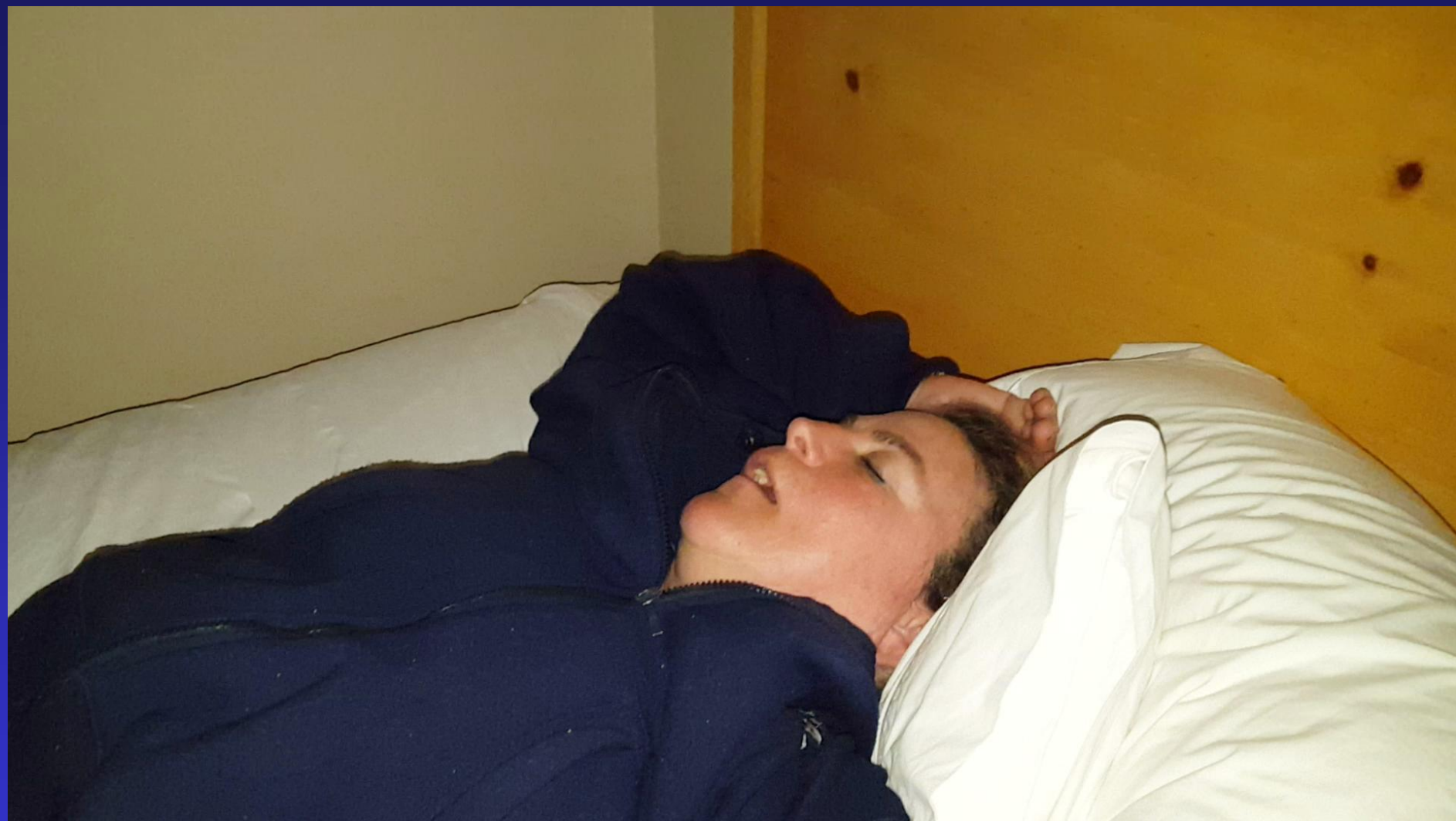
Tonsils

Normally, air flows freely past the structures in the throat.



Pediatric sleep-related breathing disorder





Sleep Apnea can kill you ≠ Natural Causes



Sleep Apnea ≠ Natural Causes



AHA SCIENTIFIC STATEMENT

Obstructive Sleep Apnea and Cardiovascular Disease: A Scientific Statement From the American Heart Association

“OSA prevalence is as high as 40% to 80% in patients with hypertension, heart failure, coronary artery disease, pulmonary hypertension, atrial fibrillation, and stroke. Despite its high prevalence in patients with heart disease and the vulnerability of cardiac patients to OSA-related stressors and adverse cardiovascular outcomes, OSA is often underrecognized and undertreated in cardiovascular practice.”

“...screening for OSA in patients with resistant/poorly controlled hypertension, pulmonary hypertension, and recurrent atrial fibrillation after either cardioversion or ablation.” “Follow-up sleep testing should be performed to assess the effectiveness of treatment.”

Categories: [Advisories & Comments](#), [Heart News](#), [Program News](#), [Stroke News](#), [Diversity/Health Equity](#) |

Published: June 29, 2022

American Heart Association adds sleep to cardiovascular health checklist

American Heart Association Presidential Advisory

Related Images



Life's Simple 7 now Life's Essential 8!

Sleep health defined as self reported hours of sleep. Ideal at 7-9 hours. “On average, how many hours of sleep do you get per night?” Consider objective sleep/actigraphy data from wearable technology, if available

Circulation

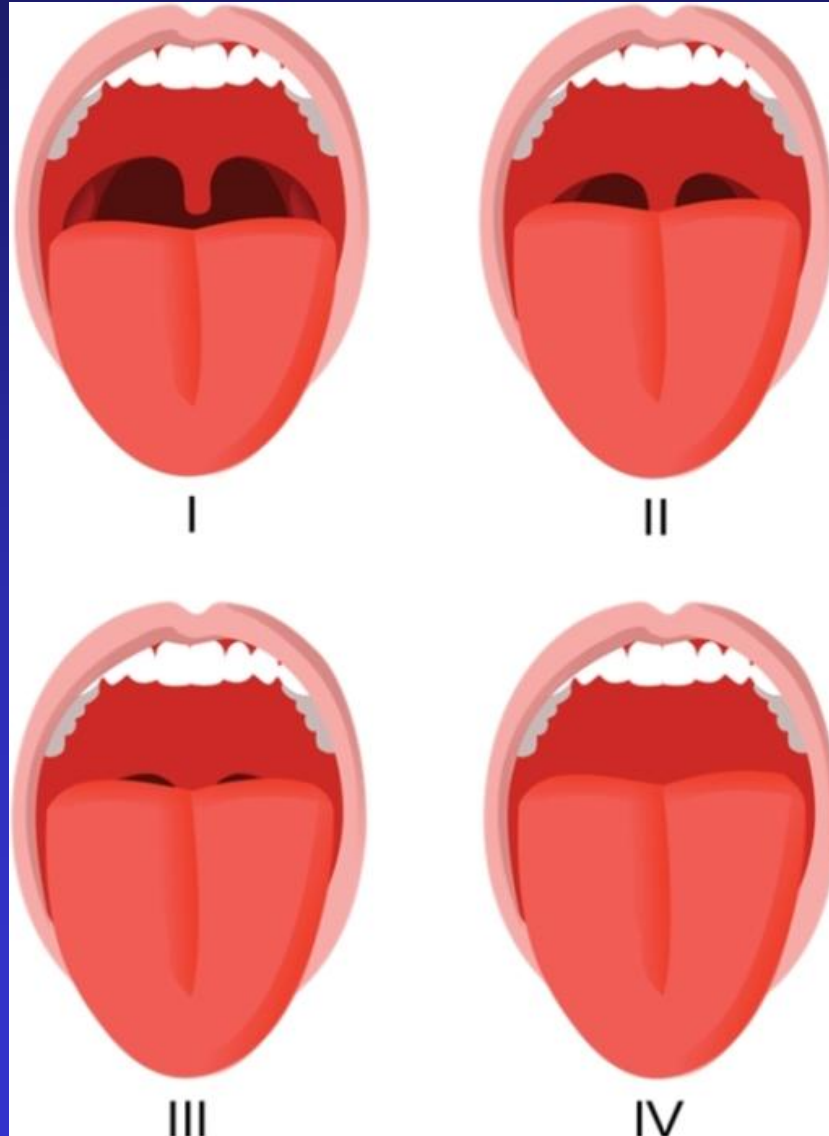
Volume 146, Issue 11, 13 September 2022; Pages 822-835
<https://doi.org/10.1161/CIRCULATIONAHA.122.060911>



ORIGINAL RESEARCH ARTICLE

Status of Cardiovascular Health in US Adults and Children Using the American Heart Association's New “Life's Essential 8” Metrics: Prevalence Estimates From the National Health and Nutrition Examination Survey (NHANES), 2013 Through 2018

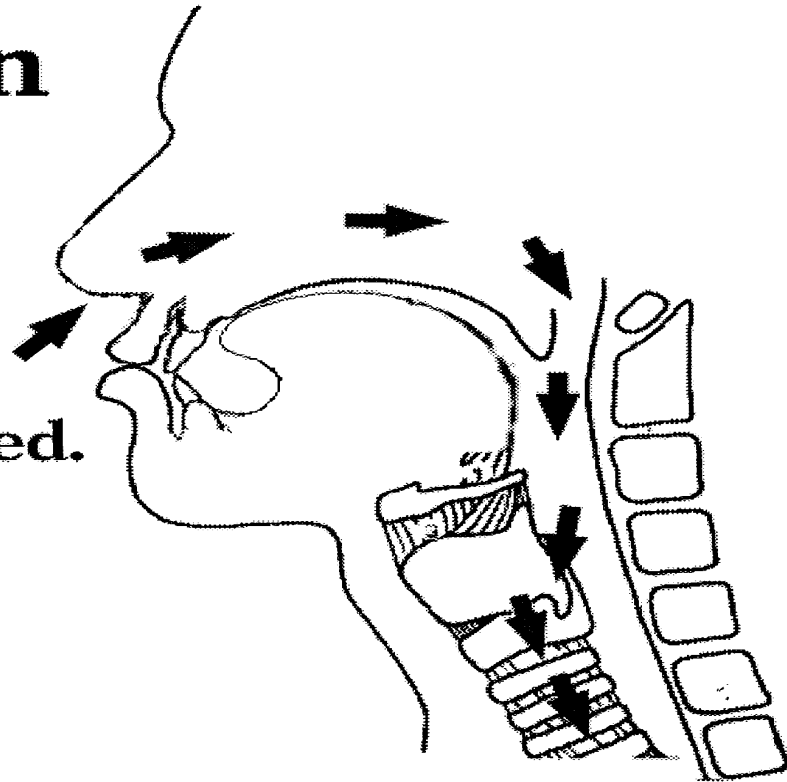
Mallampati Score



Sleep Disorders in Children

Correct Physiological Rest Position

Anterior portion of
tongue on palate.
Lips closed and relaxed.
Teeth apart.
Nasal breathing.



25:restpost-correct

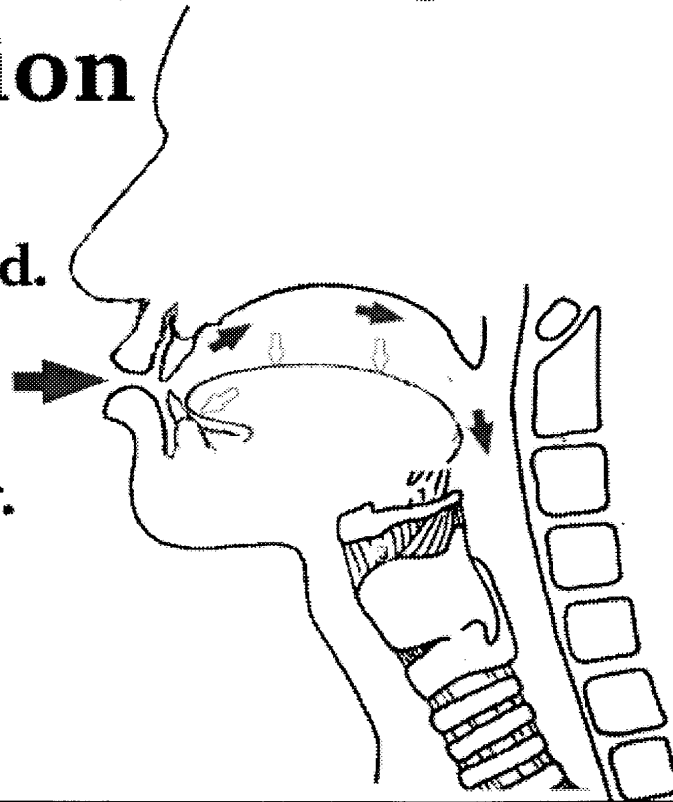
Naso-Respiratory Function and Craniofacial Growth
James McNamara 1979 as presented by James B.
DuHammell DDS



Sleep Disorders in Children

Incorrect Physiological Rest Position

Tongue depressed.
Lips apart.
Teeth apart.
Mouth breathing.
Head forward.



Naso-Respiratory Function and Craniofacial Growth
James McNamara 1979 as presented by James B.
DuHammell DDS



Mouth breathing means adenoids; adenoids mean deadened intellects.

"Mouth breathing means adenoids; adenoids mean deadened intellects." In Gulick and Ayres, *Medical inspection of schools*, 1917 (2nd ed.), p. 4.



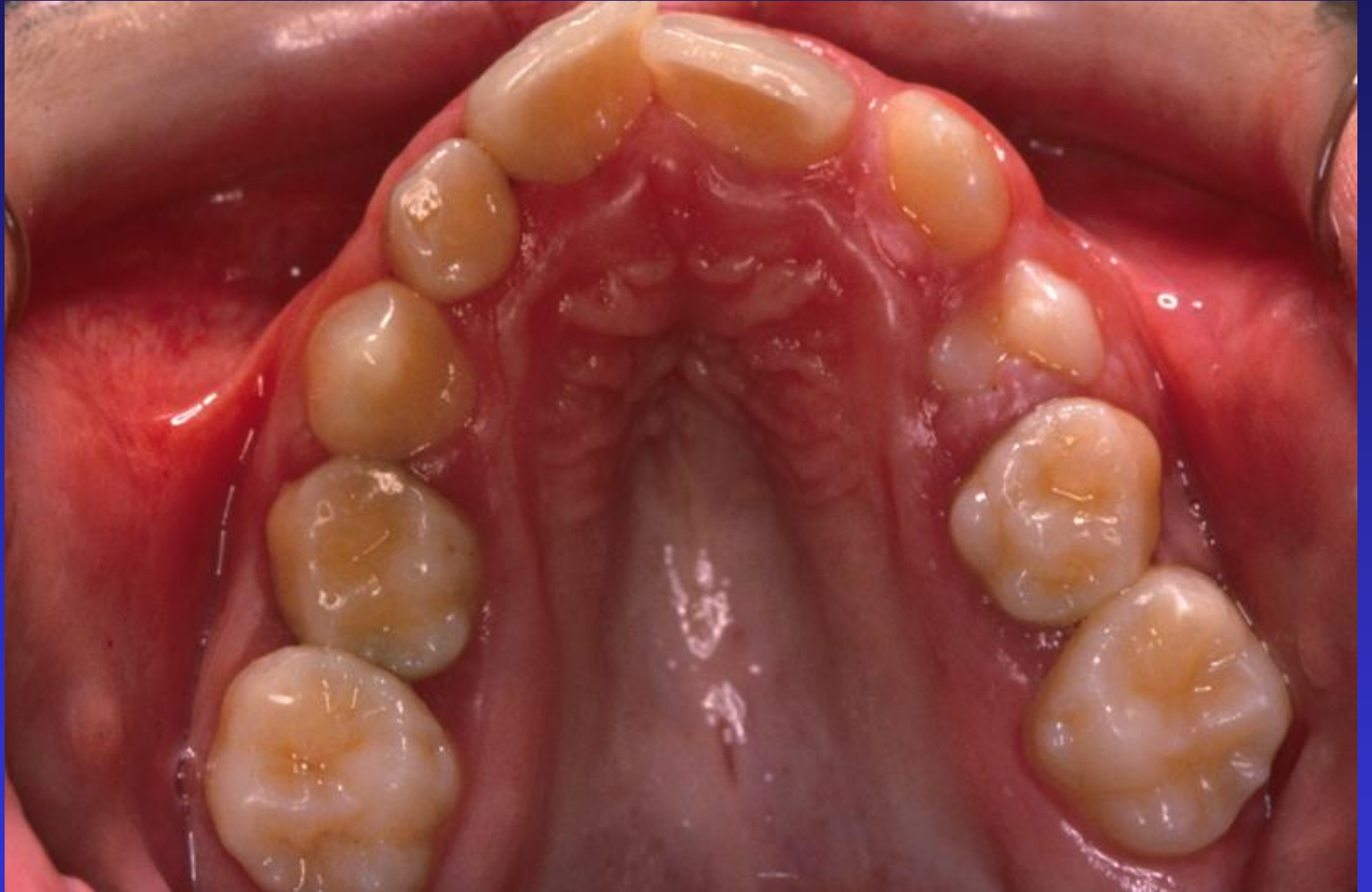
Typical adenoid faces showing mouth breathing, flattened noses, and protruding eyes.

nyamcenterforhistory.org

"Typical adenoid faces showing mouth breathing, flattened noses, and protruding eyes." In Gulick and Ayres, *Medical inspection of schools*, 1917 (2nd ed.), p. 170.

<https://nyamcenterforhistory.org/2015/03/10/adenoids-and-american-school-hygiene-in-the-early-20th-century/>





Attention-Deficit/Hyperactivity Disorder

“The essential feature of ADD/HD is a persistent pattern of inattention and/or hyperactivity-impulsivity that is more frequent and severe than is typically observed in individuals at a comparable level of development.”

“There are no specific physical features...although minor physical anomalies (...**high arched palate...**) may occur at a higher rate...” DSM IV

Snoring Predicts Hyperactivity Four Years Later Chervin 2005

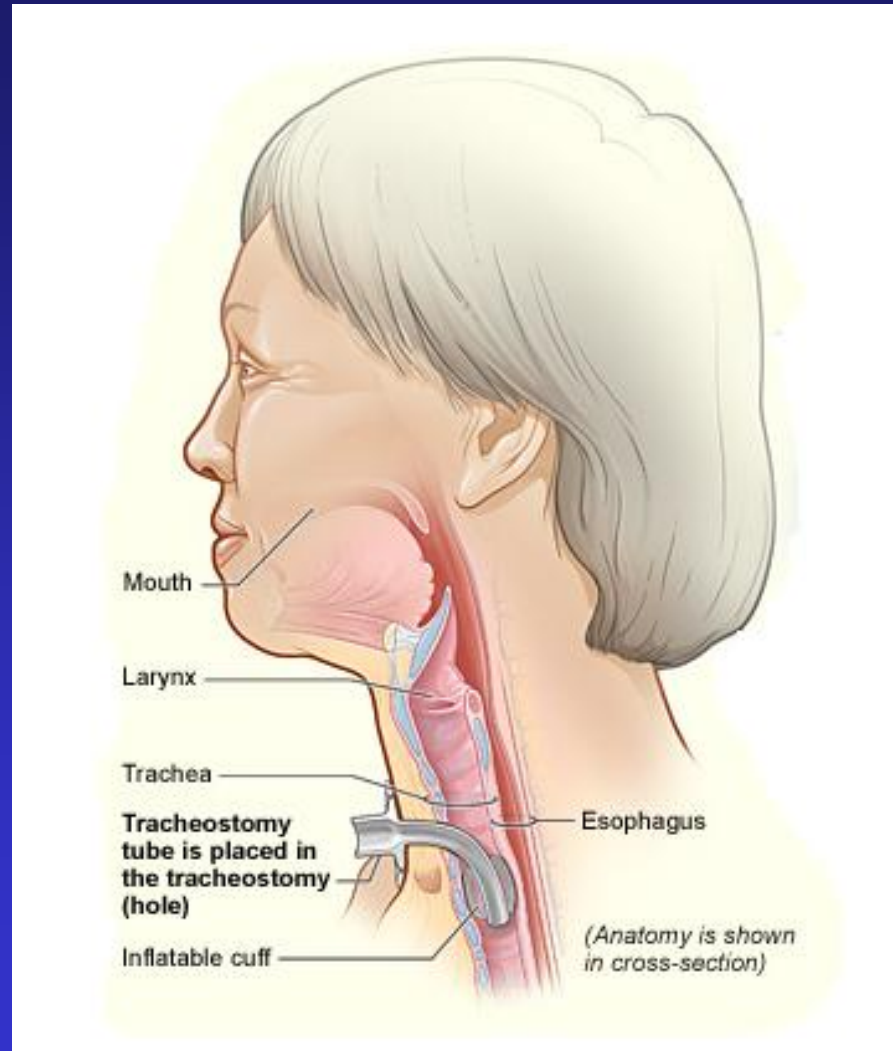
- Parents of 229 children aged 2 to 13 yrs, recruited at 2 general clinics, completed initial and 4-yr f/u surveys.
- Surveys included a Pediatric Sleep Questionnaire about snoring, sleepiness, and overall risk of SDB, and the hyperactivity index within the Conners' Parent Rating Scale.
- 13% were rated as hyperactive (hyperactivity index > 60) at f/u. Hyperactivity at f/u was predicted by baseline habitual snoring (odds ratio = 4.4) or loud snoring (4.5) and, sleepiness (3.0), or SDB (4.0).
- This 4-year prospective study shows that snoring and other symptoms of SDB are strong risk factors for future emergence or exacerbation of hyperactive behavior. These findings support the hypothesis that untreated childhood sleep-disordered breathing contributes to development of hyperactivity

OSA Treatment Options

- Conservative treatments, weight loss and positional sleep changes
- Positive airway pressure (CPAP)
- Surgery
- Orthodontics and Oral appliances
- Medications, tirzepatide



Tracheostomy was the original treatment for OSA



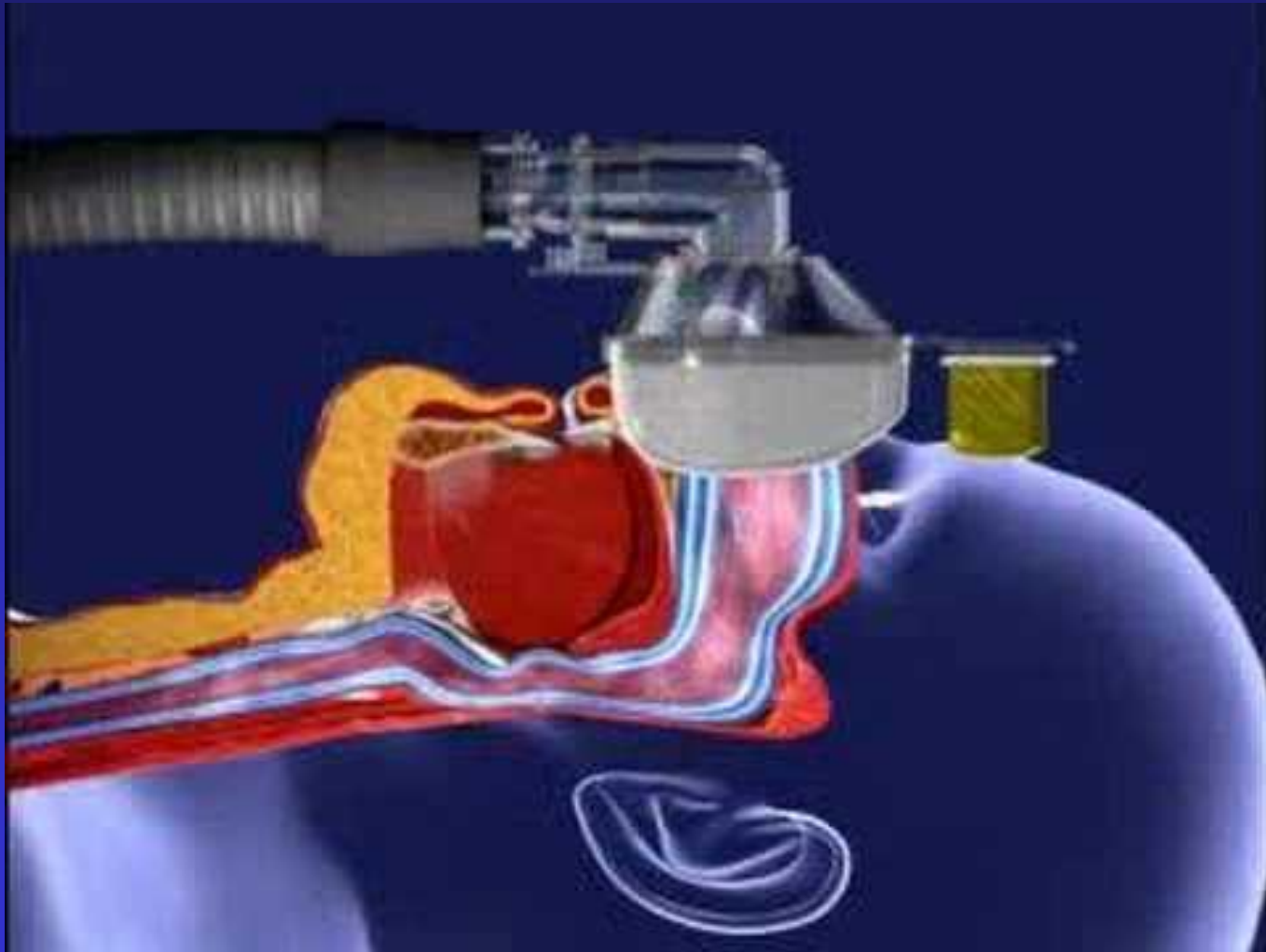
Pillows to prevent snoring



Positional Therapy



Continuous Positive Airway Pressure



Not Just plain CPAP anymore

- Auto PAP
- Bilevel
- Bilevel with a backup rate
- Auto Bilevel
- ASV (adaptive servo ventilator)
- AVAPS (average volume assured pressure support)

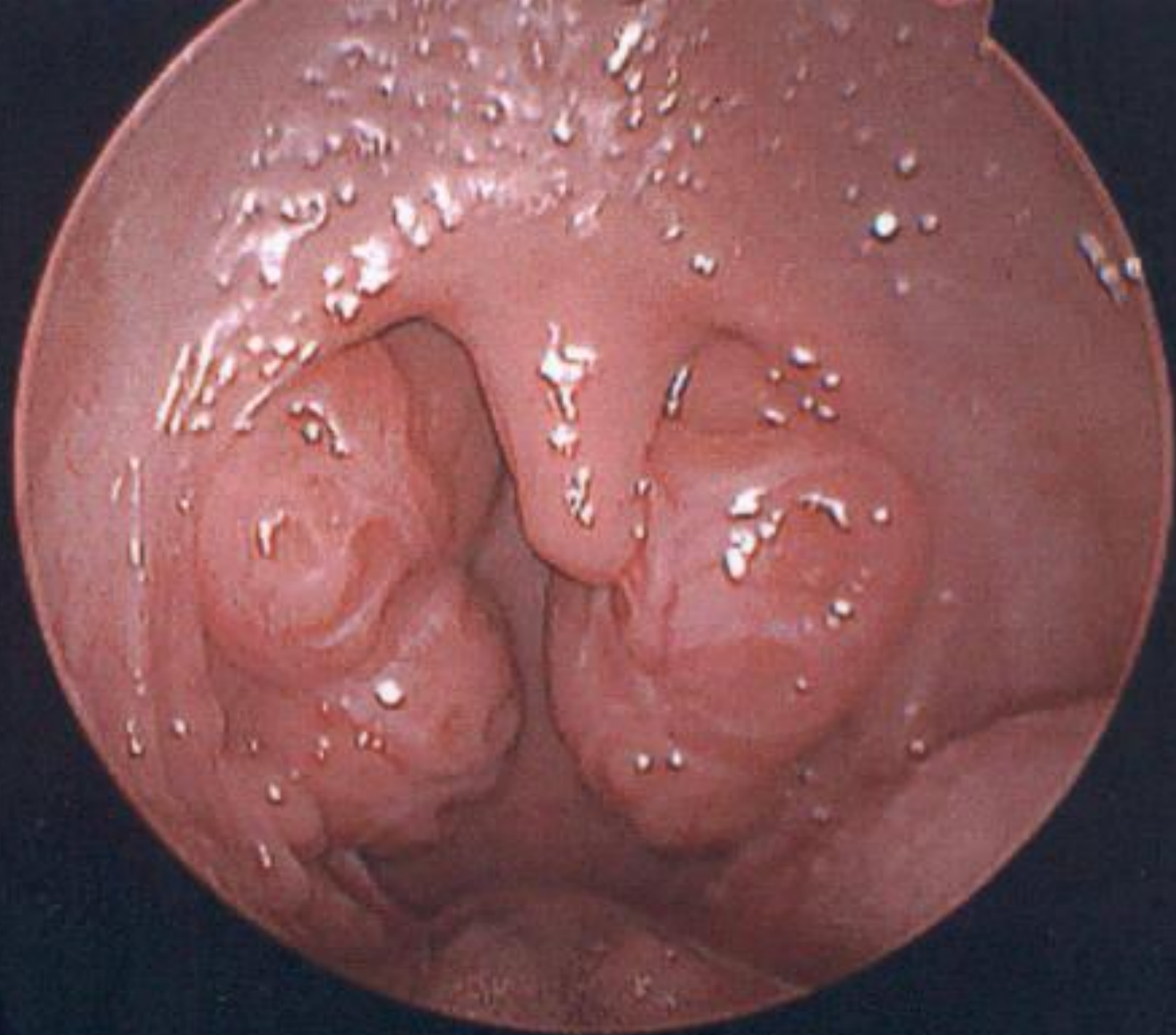
CPAP miniaturized for travel



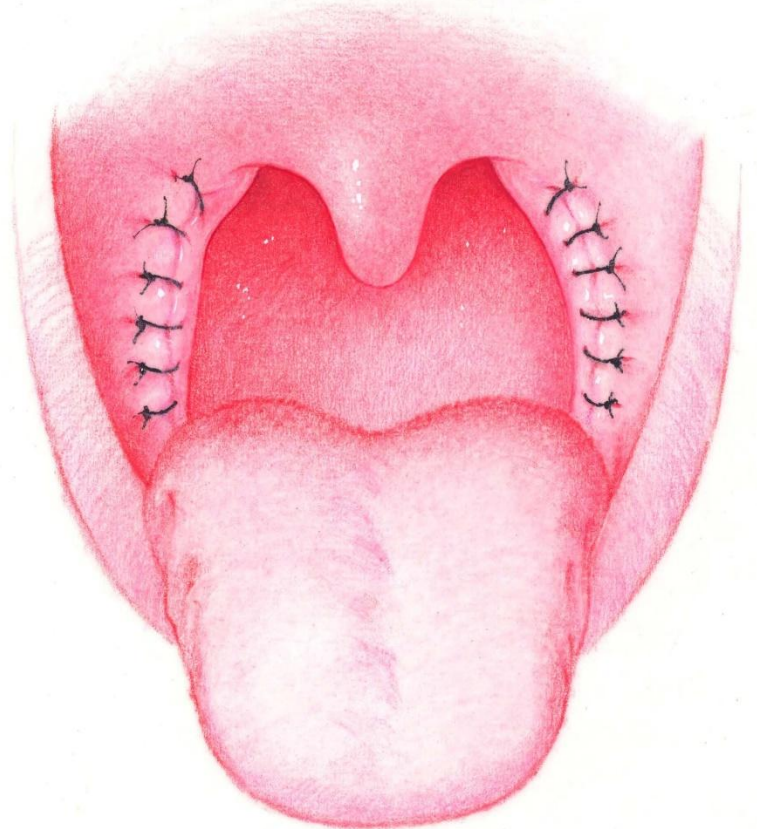
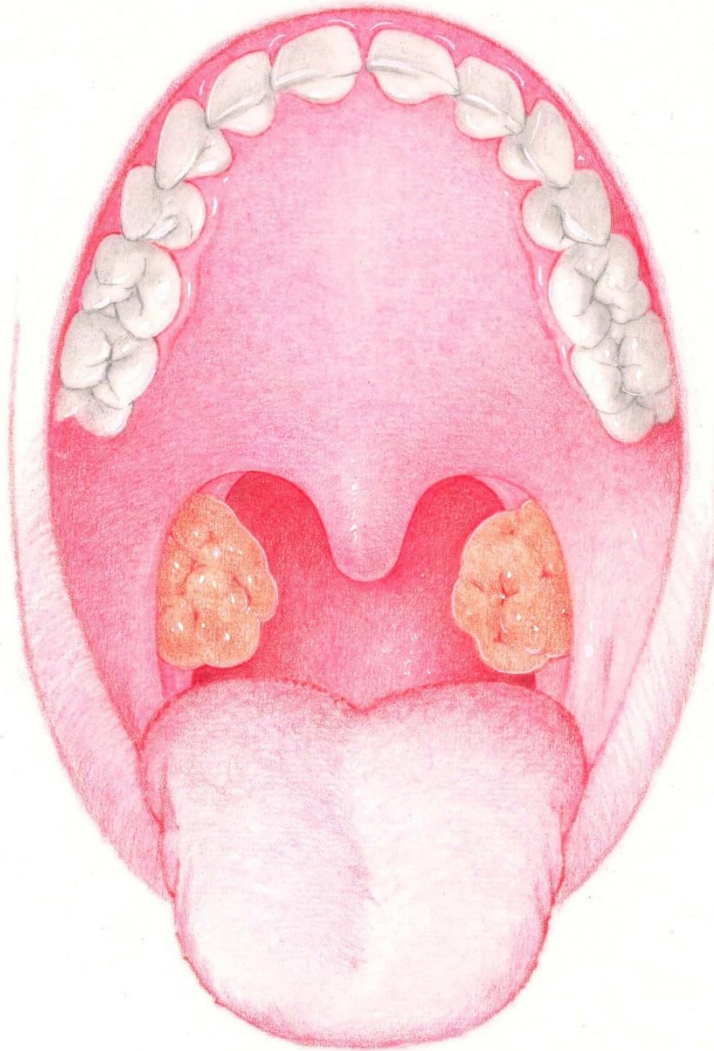
Surgery

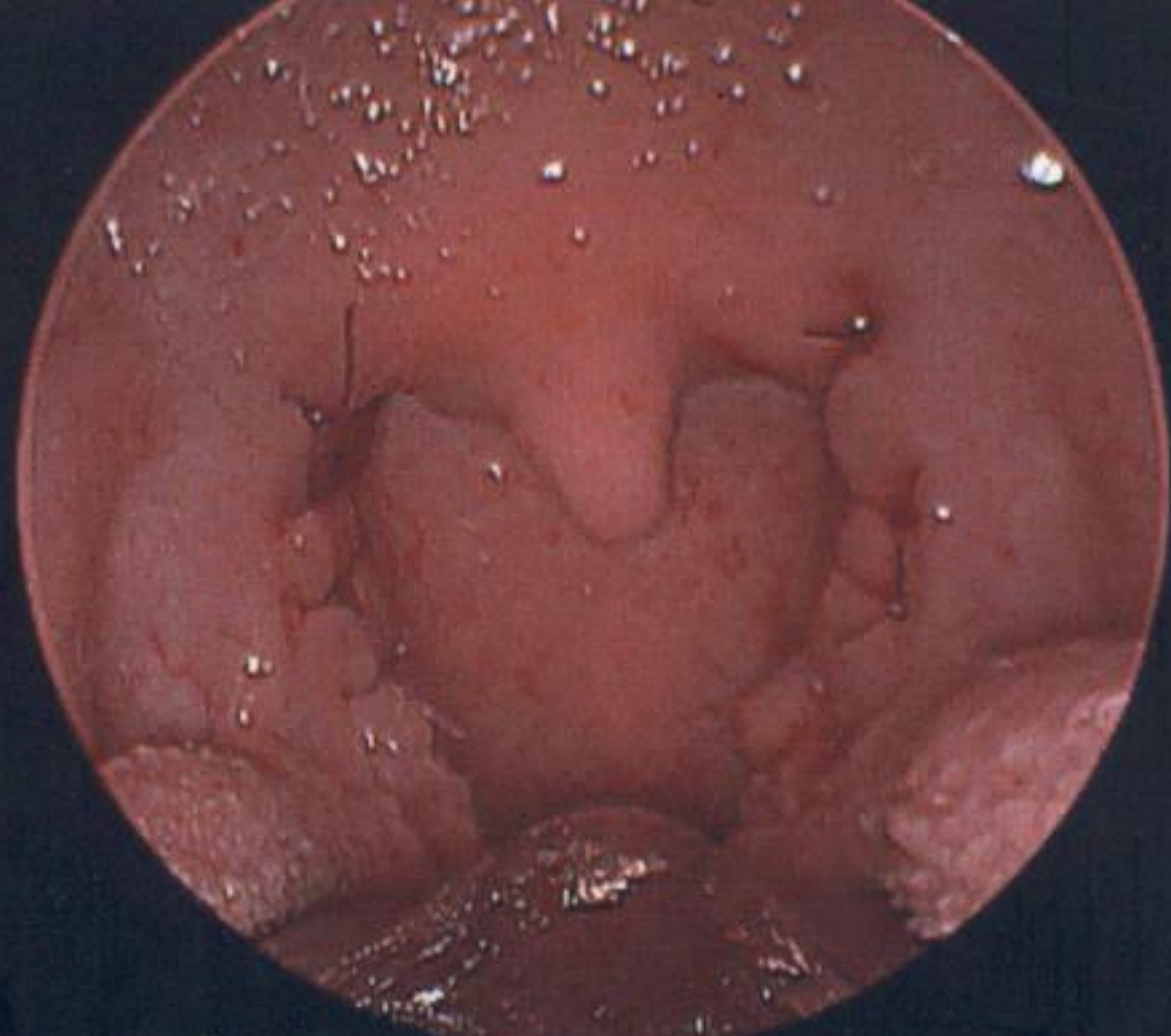
- Tonsillectomy and Adenoidectomy
- Nose Surgery
- Palate Surgery
- Jaw Surgery
- Hypoglossal Nerve Stimulators i.e.
“pacemaker” for OSA

We will have a separate lecture on sleep apnea surgery. It will have images from actual surgeries. In the past some students have felt squeamish during portions of the lecture (and some have liked it). If you are sensitive to graphic images of surgery you may want to occasionally close your eyes or if appropriate even skip class.



Adenotonsillectomy





Oral Appliance for OSA



Tirzepatide on obstructive sleep apnea-related cardiometabolic risk: secondary outcomes of the SURMOUNT-OSA randomized trial *Nature* '26

“OSA is associated with obesity and cardiovascular risk. The SURMOUNT-OSA master protocol comprised two, 52-week, randomized, double-blind, placebo-controlled phase 3 studies (study 1 and study 2) and demonstrated a significant reduction of a number of cardiometabolic risk measures in participants with OSA and obesity following treatment with tirzepatide. Here we report prespecified analysis of cardiometabolic risk measures in SURMOUNT-OSA. Post hoc analyses include changes in a homeostatic model assessment for insulin resistance and mediation analysis to determine the proportion of observed changes attributable to reductions in body weight, apnea–hypopnea index and sleep apnea-specific hypoxic burden. In both study 1 and study 2 of SURMOUNT-OSA, tirzepatide treatment was associated with greater alleviation of cardiometabolic risk factors than placebo. Independent mediation effect of changes in OSA metrics was observed on high-sensitivity C-reactive protein, homeostatic model assessment for insulin resistance and triglycerides. The combination of changes in weight and OSA metrics, as well as weight alone, had a significant mediation effect on systolic blood pressure, but there was no significant mediation effect of weight or OSA metrics observed on diastolic blood pressure. Based on the mediation analysis, treating both sleep-disordered breathing and obesity is likely required to optimize the treatment effect on cardiometabolic benefits for patients with moderate-to-severe OSA and obesity”

Obstructive Sleep Apnea

- Brain prioritizes sleeping over breathing!
- Snoring is never normal, but it is common
- Independent cardiovascular risk factor
- OSA can occur at any age, gender or BMI
- Treatable and curable

Questions?

